

Day 1

Beyond the Syntax: Mastering Agentic Coding

The Future of Software Engineering

Awwal Badru

B.Ed CS/Maths (Nigeria) | M.Sc CS (Australia) |

MS, CS & PhD Candidate (United States)

Areas of specialization:

Empirical Software Engineering; Software Security

Software Engineering Education; Computer Science Education



VS



VS



Workshop Outline

DAY 1: Foundations & Demo

- The Evolution of AI & Coding
- Coding Agents & Agentic Loop
- Demo: Context, Action & Verification
- Interactive Q & A
- IDE Setup and Resources

Workshop Outline

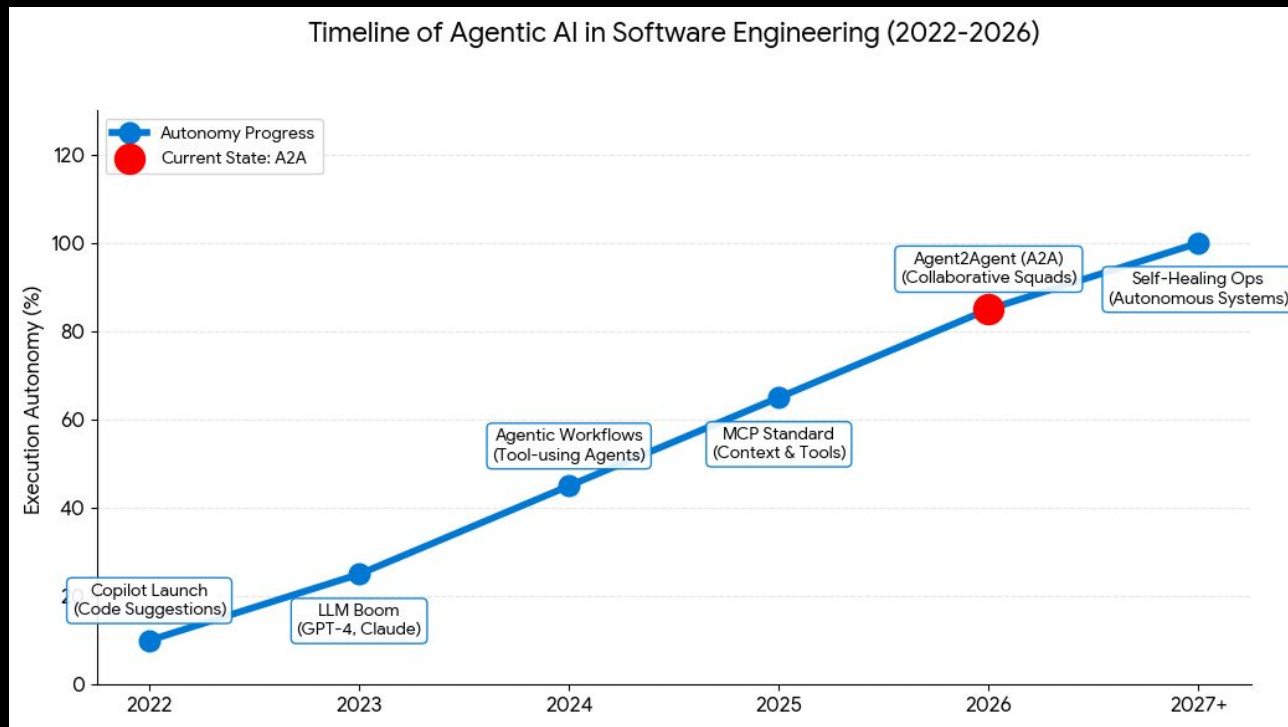
DAY 1: Foundations & Demo

- The Evolution of AI & Coding
- Coding Agents & Agentic Loop
- Demo: Context, Action & Verification
- Interactive Q & A
- IDE Setup and Resources

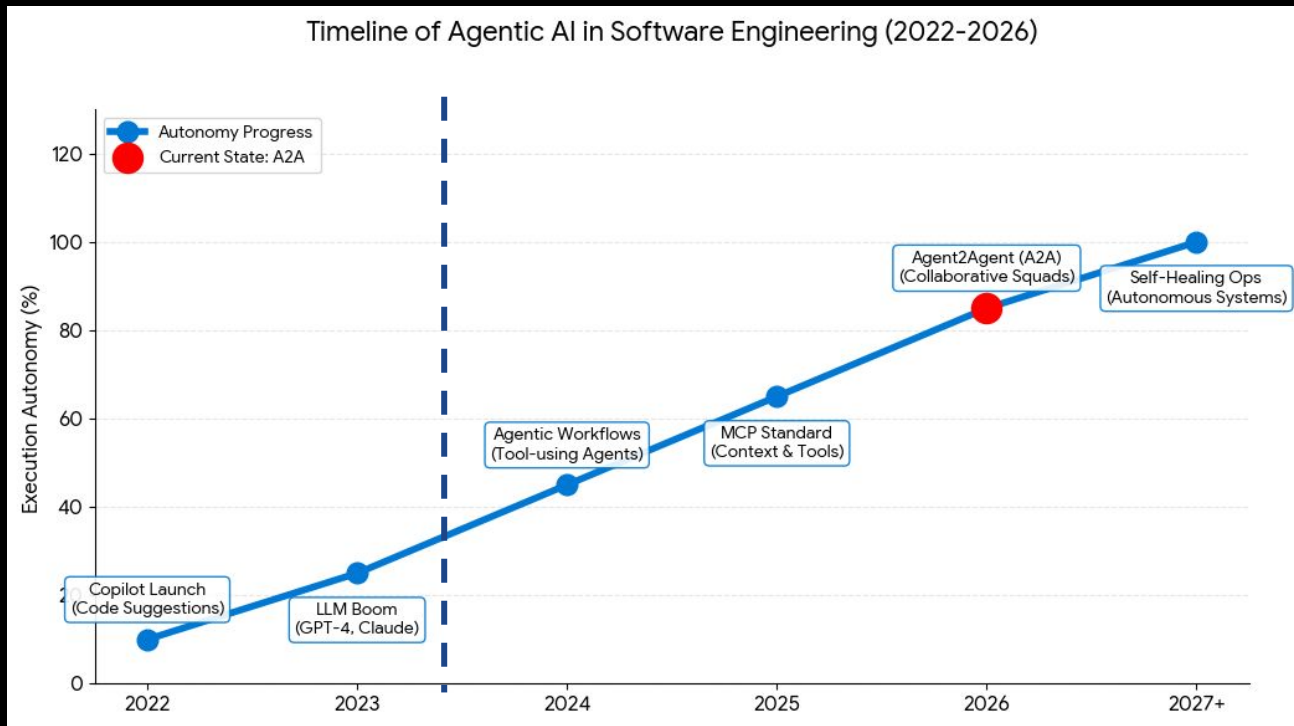
DAY 2: Practical Application

- Software Development Life Cycle
- Hands-on Building with Agents
- Git Versioning & Collaboration

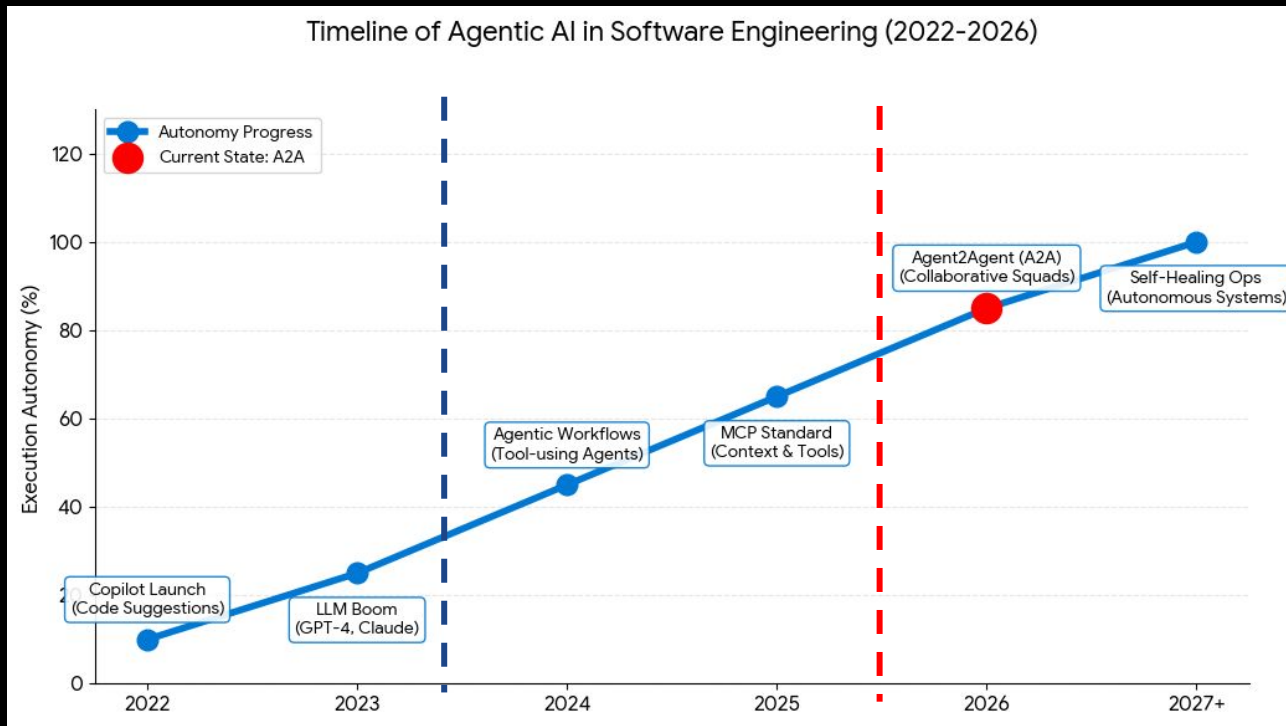
The Evolution of AI & Coding



The Evolution of AI & Coding



The Evolution of AI & Coding



Why Mastering Agentic AI?

AI · CODING

**An AI-powered coding tool
wiped out a software
company's database, then
apologized for a
'catastrophic failure on my
part'**

BY BEATRICE NOLAN
TECH REPORTER

July 23, 2025 at 7:22 AM EDT



Why Mastering Agentic AI?

AI · CODING

An AI-powered coding tool wiped out a software company's database, then apologized for a 'catastrophic failure on my part'

BY BEATRICE NOLAN
TECH REPORTER

July 23, 2025 at 7:22 AM EDT



Amjad Masad  
@amasad

We saw Jason's post. @Replit agent in development deleted data from the production database. Unacceptable and should never be possible.

- Working around the weekend, we started rolling out automatic DB dev/prod separation to prevent this categorically. Staging environments in the works, too. More tomorrow.

Why Mastering Agentic AI?

It took 9 seconds: AI agent running on Anthropic's Claude Opus 4.6 wipes critical database

Story by Business Today Desk • 19h • ⌚ 3 min read

Source:

<https://www.msn.com/en-in/money/topstories/it-took-9-seconds-ai-agent-running-on-anthropic-s-claude-opus-4-6-wipe-s-critical-database/ar-AA21Np4C?ocid=socialshare>

The Evolution of AI & Coding

1

Code Completion

Intelligent snippets and autocomplete based on context. Tools like traditional IDE intellisense.



The screenshot shows a code editor with a file named 'hello_world.py'. The code is as follows:

```
1 def hello_world():  
2     print("Hello, World!")  
3  
4 if   
5     hello_world()
```

A red box highlights the 'hello_world()' suggestion in the autocomplete dropdown menu. The dropdown menu also shows 'Accept Word' and 'Accept' buttons.

The Evolution of AI & Coding

2

Code Generation

Generative AI creating entire functions and boilerplate from natural language prompts. The era of Copilots.

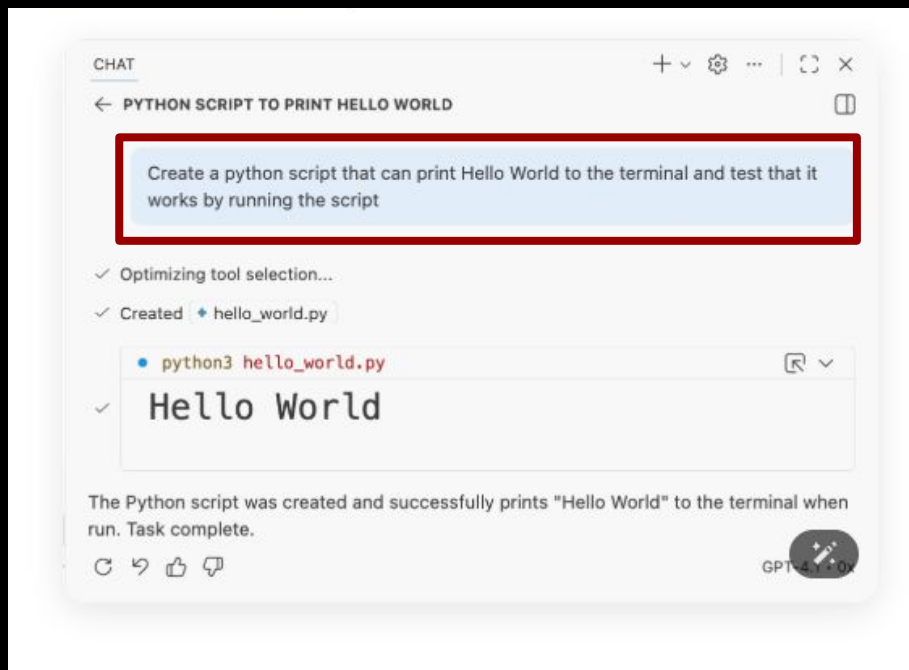


The Evolution of AI & Coding

3

Agentic Coding

Autonomous agents that reason, plan, and execute across files, using tools and verifying results.



Coding Agents & Demonstration

IDE

- Copilot in VS Code
- Cursor
- Windsurf
- Antigravity

Coding Agents & Demonstration

IDE

- Copilot in VS Code
- Cursor
- Windsurf
- Antigravity

CLI

- Copilot CLI
- Claude Code
- Pi

Coding Agents & Demonstration

IDE

- Copilot in VS Code
- Cursor
- Windsurf
- Antigravity

CLI

- Copilot CLI
- Claude Code
- Pi

Web

- Copilot in GitHub.com
- Codex
- Jules

Coding Agents & Demonstration



IDE

- Copilot in VS Code
- Cursor
- Windsurf
- Antigravity



CLI

- Copilot CLI
- Claude Code
- Pi



Web

- Copilot in GitHub.com
- Codex
- Jules



Custom

- Copilot SDK
- PyDantic AI
- LangChain

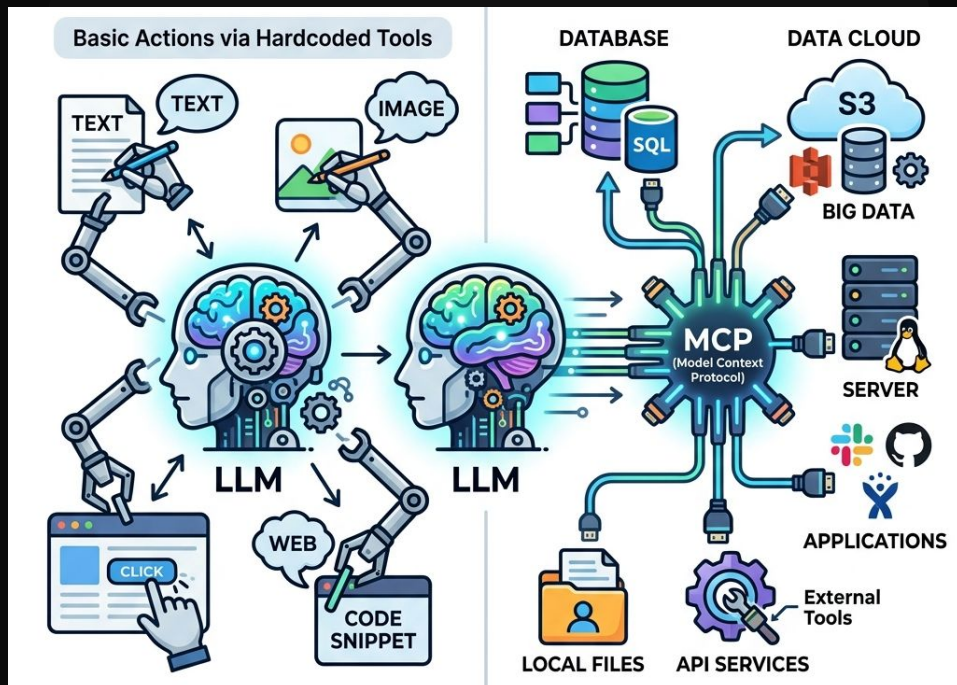
What is an AI Agent?

An AI Agent is an LLM that calls tools in a loop to achieve a goal.

Source: https://simonwillison.net/2025/Sep/18/agents/?utm_source=chatgpt.com

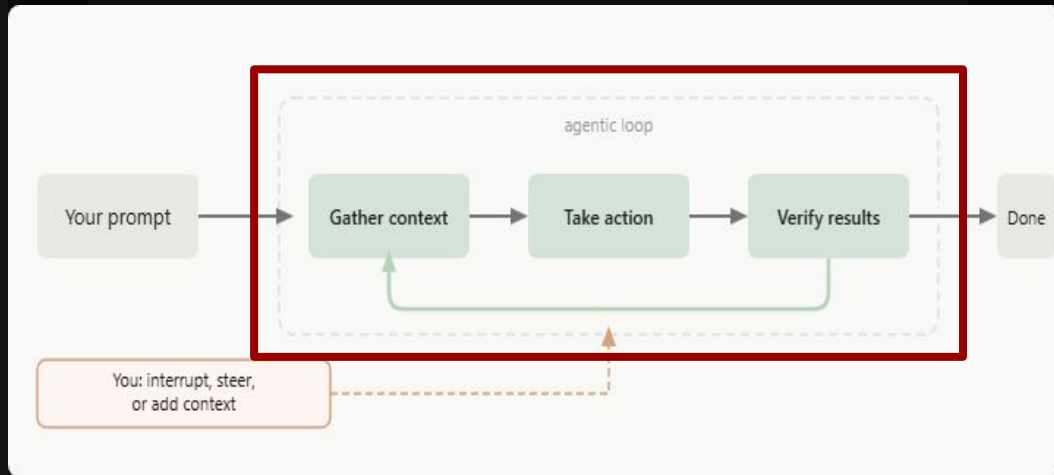
What is an AI Agent?

An AI Agent is an **LLM that calls tools** in a loop to achieve a goal.



What is an AI Agent?

An AI Agent is an **LLM that calls tools** in a **loop** to achieve a goal.



Source: <https://code.claude.com/docs/en/how-claude-code-works>

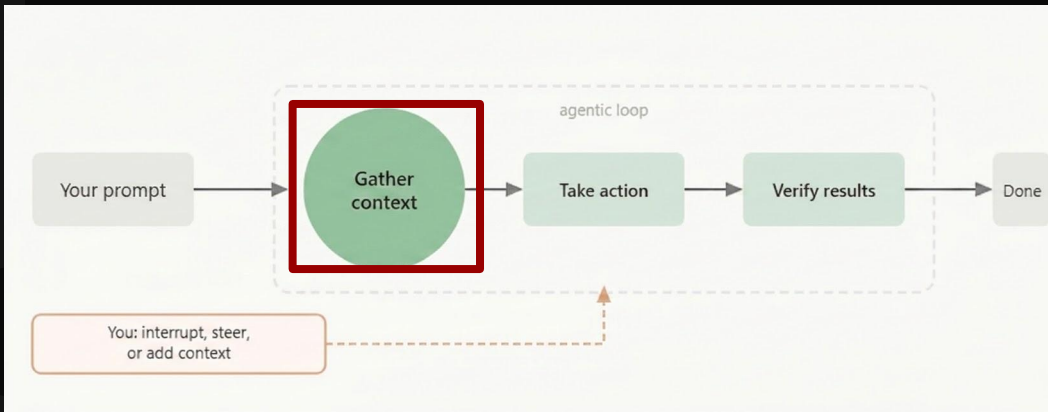
Quick Pricing Overview (April 2026)

Service	Individual Plan	High-Usage/Max	Team/Business
GitHub Copilot	\$10 /mo	\$39 /mo (Pro+)	\$19 /user/mo
Claude	\$20 /mo	\$100 - \$200 /mo	\$25 /user/mo
Antigravity	\$29 /mo	\$99 - \$250 /mo	Custom
OpenAI Codex	\$20 /mo (via ChatGPT Plus)	\$200 /mo (Pro)	\$25 /user/mo

Equipping Agents with Context

Including Files and Assets

- Directly provide files, images, and GitHub items (Issues, PRs)
- Offer exemplar files containing best-practice code



Directive Documentation

- `copilot-instructions.md` — GitHub Copilot
- `cursor rules` — Specific to Cursor IDE
- `claude.md` — Tailored for Claude Code
- `agents.md` — The standard OpenAI format

Providing Tools to an Agent

MCP — Model Context Protocol

Open protocol for giving agents access to tools

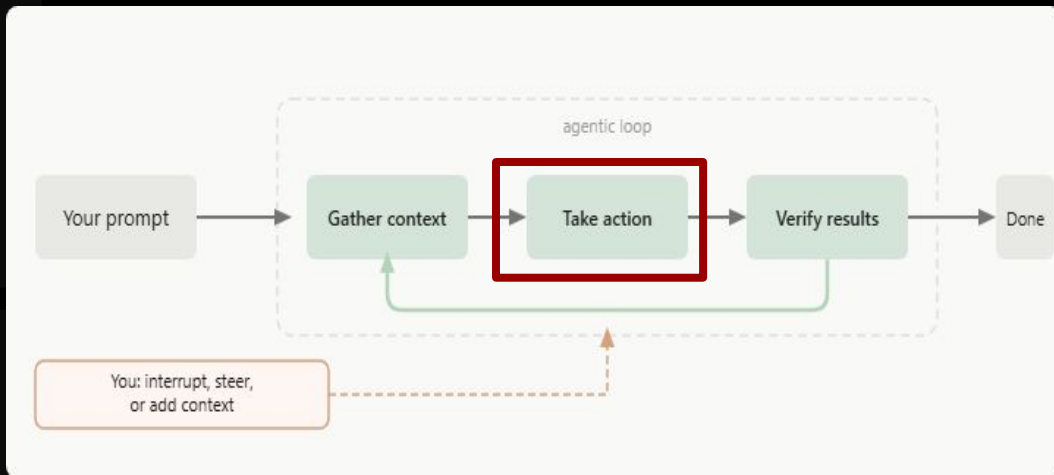
In IDE: search @mcp in extensions for vetted servers

Pro Tip: You can also create your own MCP servers

Agent Skills

A reusable workflow defined in a `SKILL.MD` file

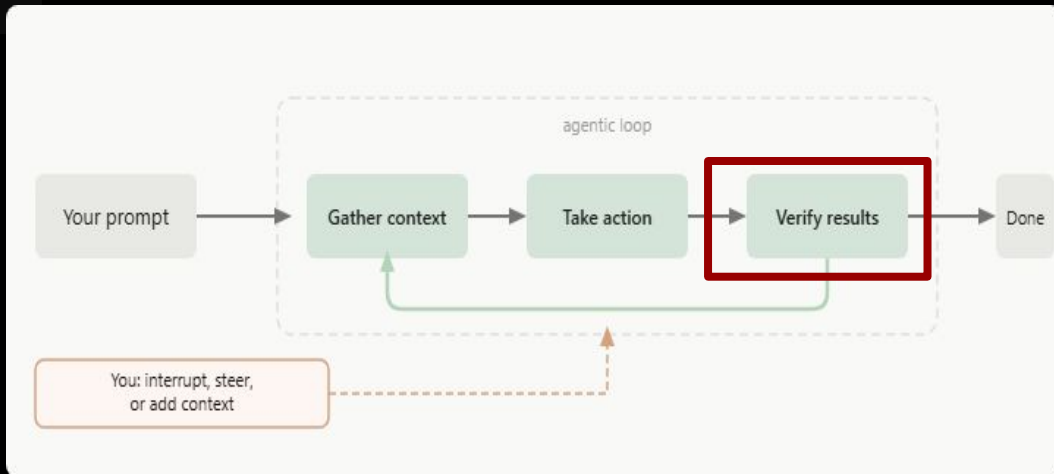
Skills can include tools, MCP servers, scripts, and more!



Verification Plan

Automated Tests - Playwright

Manual Verification



Resources

IDE (VS Code)

- VS Code Installation
<https://code.visualstudio.com/download>
- Copilot Installation from VS Code extension
<https://www.youtube.com/watch?v=ws6t0NBle9g>
- Launching Copilot for free (please you need to have a github account)
https://www.youtube.com/watch?v=X_Aet9ndh_Y

IDE (Antigravity)

- Google Agent IDE Installation
<https://antigravity.google/download>

You need to have gmail account to sign in and start using the agent for free

Resources

CLI (Copilot)

- Installation npm (requires Node.js 22+)

<https://www.youtube.com/watch?v=BDxRhhs36ns>

- Installation on windows via (WinGet)

https://www.youtube.com/watch?v=3mCaZ_tYdI0

CLI (Copilot)

Install commands:

Windows (WinGet): winget install
GitHub.Copilot

npm (requires Node.js 22+): npm install
-g @github/copilot

macOS / Linux: brew install copilot-cli

Installation Support



Peer Support

Meet your colleagues for troubleshooting and technical guidance

OR



WhatsApp Support

Reach out to our dedicated support team on WhatsApp

Niyi: 0803-410-6706

Tossyno: 0703-448-6740

Prof. Dami: 0803-945-3555

DLaw: 0706-788-5378

Analyst: 0703-844-8548

Day 2

Beyond the Syntax: Mastering Agentic Coding

The Future of Software Engineering

Awwal Badru

B.Ed CS/Maths (Nigeria) | M.Sc CS (Australia) |

MS, CS & PhD Candidate (United States)

Areas of specialization:

Empirical Software Engineering; Software Security

Software Engineering Education; Computer Science Education



VS



VS



Workshop Outline

DAY 1: Foundations & Demo

- The Evolution of AI & Coding
- Coding Agents & Agentic Loop
- Demo: Context, Action & Verification
- IDE Setup and Resources
- Interactive Q & A

DAY 2: Practical Application

- Software Development Life Cycle
- Hands-on Building with Agents
- Git Versioning & Collaboration

SDLC

The 7 Phases of SDLC

1. Planning

Define scope, costs, and identify risks.

SDLC

The 7 Phases of SDLC

1. Planning

Define scope, costs, and identify risks.

2. Requirements

Gather needs from stakeholders (the "what").

SDLC

The 7 Phases of SDLC

1. Planning

Define scope, costs, and identify risks.

2. Requirements

Gather needs from stakeholders (the "what").

3. Design

Architecture, UI/UX, and data models.

SDLC

The 7 Phases of SDLC

1. Planning

Define scope, costs, and identify risks.

2. Requirements

Gather needs from stakeholders (the "what").

3. Design

Architecture, UI/UX, and data models.

4. Implementation

The actual writing of the source code.

SDLC

The 7 Phases of SDLC

1. Planning

Define scope, costs, and identify risks.

2. Requirements

Gather needs from stakeholders (the "what").

3. Design

Architecture, UI/UX, and data models.

4. Implementation

The actual writing of the source code.



5. Testing

Verify software for bugs and requirements.

SDLC

The 7 Phases of SDLC

1. Planning

Define scope, costs, and identify risks.

2. Requirements

Gather needs from stakeholders (the "what").

3. Design

Architecture, UI/UX, and data models.

4. Implementation

The actual writing of the source code.



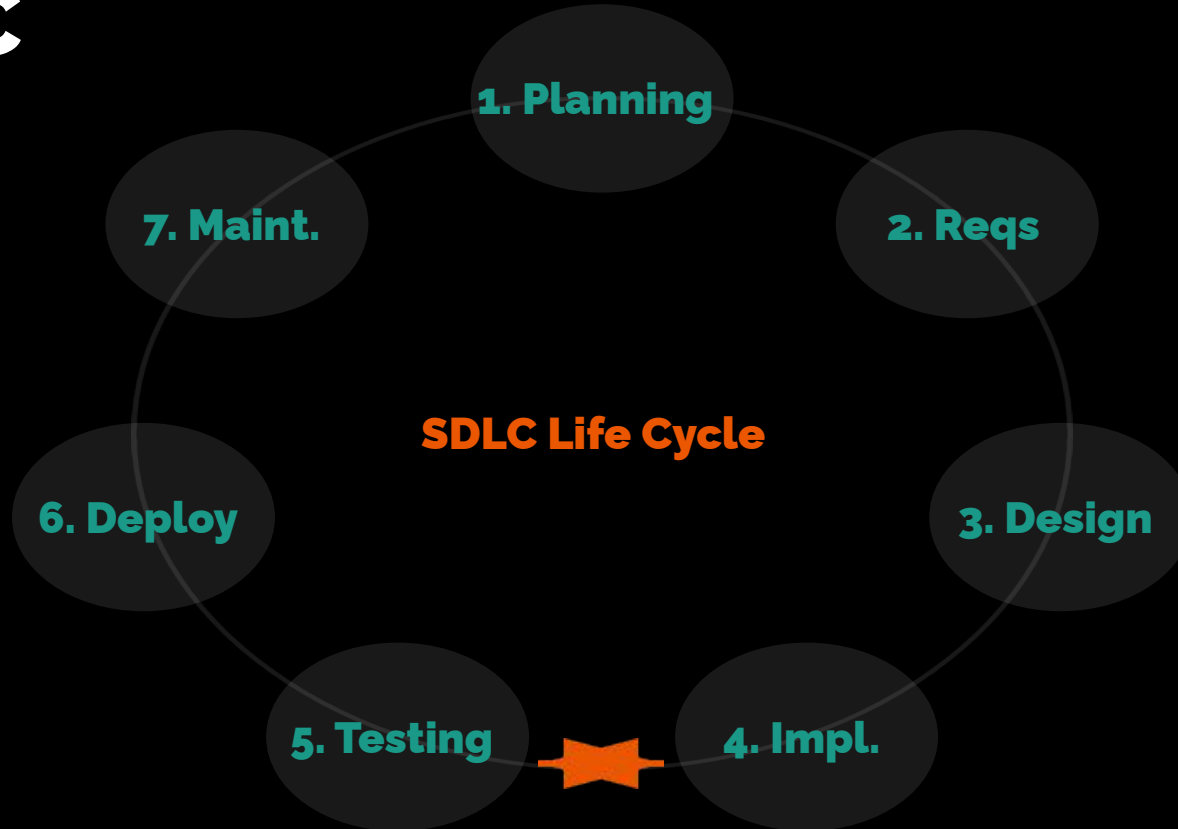
5. Testing

Verify software for bugs and requirements.

6. Deployment

Release to production environment.

SDLC



Hands-on Learning

- For UX/UI

<https://stitch.withgoogle.com/>

Hands-on Learning

What app are you building? It's time to build your desire app

1. Planning

Define scope, costs, and identify risks.

2. Requirements

Gather needs from stakeholders (the "what").

3. Design

Architecture, UI/UX, and data models.

4. Implementation

The actual writing of the source code.

5. Testing

Verify software for bugs and requirements.